

ServiceNow – SAP S/4HANA OData Integration

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Scenarios

- In this scenario we will execute a ServiceNow OData workflow that will connect to S/4HANA OData interface and fetch suppliers and modify a supplier.

Assumptions

- Attached ServiceNow Update set is loaded in the ServiceNow instance.
- Infrastructure: SAP S4/HANA OData interface, ServiceNow Integration Hub, and SAP S/4HANA are all set-up.

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SAP Configuration (S/4HANA OData)

1. Pre-requisites: Activate all the OData API Resources in SAP S/4 HANA
2. Use the T-code SICF.
3. Give the required service name and click on Activate Service

In our case we have tested LOOK UP VENDOR below the OData URI for the same.

- Look up Business Partner Addresses Stream
/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_BusinessPartnerAddress

One has to activate the necessary OData Services in SICF which will be called from ServiceNow.

ServiceNow Integration Hub's SAP S4/HANA OData Spoke Configuration

Purpose: In the previous section, the SAP S/4HANA has been configured. Core S/4HANA OData spoke actions will be used to look up supplier names and modify a supplier. In this section, we will go over how to set up ServiceNow's SAP S/4HANA OData Spoke.

Step 1 – ServiceNow: Create Credential Object

1. Provide Username and password for OData endpoint connection. (this will be provided by SAP team)

The screenshot shows the ServiceNow interface for creating a credential object. The top section is titled "Basic Auth Credentials OData Spoke Credential" and includes "Update" and "Delete" buttons. The form fields are as follows:

- Name:** OData Spoke Credential
- Order:** 100
- User name:** RFC_USER
- Password:** (masked with dots)
- Active:** ☒

Below the form is a blue informational box: "A connection alias resolves your connection and credential at runtime. More than one Credential can be active per Connection Alias at a time. If more than one credential is active, they will be used in order." Below this is a "Credential alias" field with a lock icon.

The bottom section is titled "Discovery IP Affinity" and includes a "MID server" dropdown, a "Search" field, and a "New" button. Below this is a table titled "Credential Affinities":

	MID server	Credential ID	IP address	IP address decimal	Domain	Type
No records to display						

Step 2 – ServiceNow: Create Connection in SAP_S4HANA_OData connection alias

1. Go to Connection & Credential Aliases
2. Open alias SAP_S4HANA_OData
3. Create new connection as following: choose credentials created in previous step, provide host that serves as endpoint for OData connection, check use mid server option.

The screenshot displays the 'HTTP(s) Connection' configuration page for an 'OData Spoke Connection'. The interface includes a top navigation bar with a back arrow, a menu icon, the title 'HTTP(s) Connection OData Spoke Connection', and action buttons for 'Update', 'Delete', and navigation arrows. The main configuration area contains several fields: 'Name' (OData Spoke Connection), 'Active' (checked), 'Credential' (OData Spoke Credential), 'Domain' (global), and 'Connection alias' (sn_hana_odata_spk.SAP_S4_HANA_). A blue informational banner states: 'A connection alias resolves your connection and credential at runtime. Only one Connection is active per Connection Alias at a time. [More Info](#)'. Below this, another blue banner says: 'Manually enter your connection URL or use the URL builder to build the connection string. [More Info](#)'. The 'URL builder' checkbox is unchecked. The 'Connection URL' field contains 'http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000'. The 'Use MID server' checkbox is checked. The 'Connection timeout' is set to 0. A section titled 'Advanced MID Server Configuration' is expanded, showing 'MID Selection' set to 'Auto-Select MID Server'. A blue banner below this section reads: 'This is an advanced feature for System Administrators to narrow down MID Server Selection Criterion. [More Info](#)'. The 'Capabilities' section is locked, indicated by a padlock icon. A final blue banner states: 'MID Applications are a higher level grouping beyond Capabilities. Use 'ALL' by default. [More Info](#)'. The 'MID Application' dropdown is set to 'ALL'.

HTTP(s) Connection
OData Spoke Connection

* Name: OData Spoke Connection

Credential: OData Spoke Credential

Active: ☒

Domain: global

A connection alias resolves your connection and credential at runtime. Only one Connection is active per Connection Alias at a time. [More Info](#)

* Connection alias: sn_hana_odata_spk.SAP_S4_HANA_

Manually enter your connection URL or use the URL builder to build the connection string. [More Info](#)

URL builder: ☐

* Connection URL: http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000

Use MID server: ☒

Connection timeout: 0

Advanced MID Server Configuration

MID Selection: Auto-Select MID Server

This is an advanced feature for System Administrators to narrow down MID Server Selection Criterion. [More Info](#)

Capabilities:

MID Applications are a higher level grouping beyond Capabilities. Use 'ALL' by default. [More Info](#)

MID Application: ALL

Step 3 – ServiceNow: Enable Connection (optional)

If there are multiple connections under the same alias, make sure that newly created connection is active.

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Connection & Credential Aliases

sn_hana_odata_spk.SAP_S4_HANA_OData

⚙

...

↑

↓

⚠

This record is in the [SAP S4 HANA OData Spoke](#) application, but [Global](#) is the current application. To edit this record click [here](#).

Name

SAP S4 HANA OData

ID

sn_hana_odata_spk.SAP_S4_HANA_ODat

Application

SAP S4 HANA OData Spoke

①

Type

Connection and Credential

Connection type

HTTP

Support Multiple Active Connections

☐

Default Retry Policy

Default HTTP Retry Policy

Configuration Template

OData Configuration Template

Related Links

[Create New Connection & Credential](#)

Connections (3)

Connection Attributes

Child Aliases

≡

🔍

Name

▼

Search

⚙

—

Actions on selected rows...

▼

New

Connection alias = sn_hana_odata_spk.SAP_S4_HANA_OData

☐	🔍	Name ▲	Connection type	Active	Credential	Use MID server	Protocol	Host	Override default port
☐		OData BTP	HTTP(s) Connection	false	OData BTP	true	https	integration-suite-vduzfeoz.it-cpi024-rt...	
☐		OData Spoke Connection	HTTP(s) Connection	true	OData Spoke Credential	true	https	ec2-18-170-34-13.eu-west-2.compute.amazo...	8000

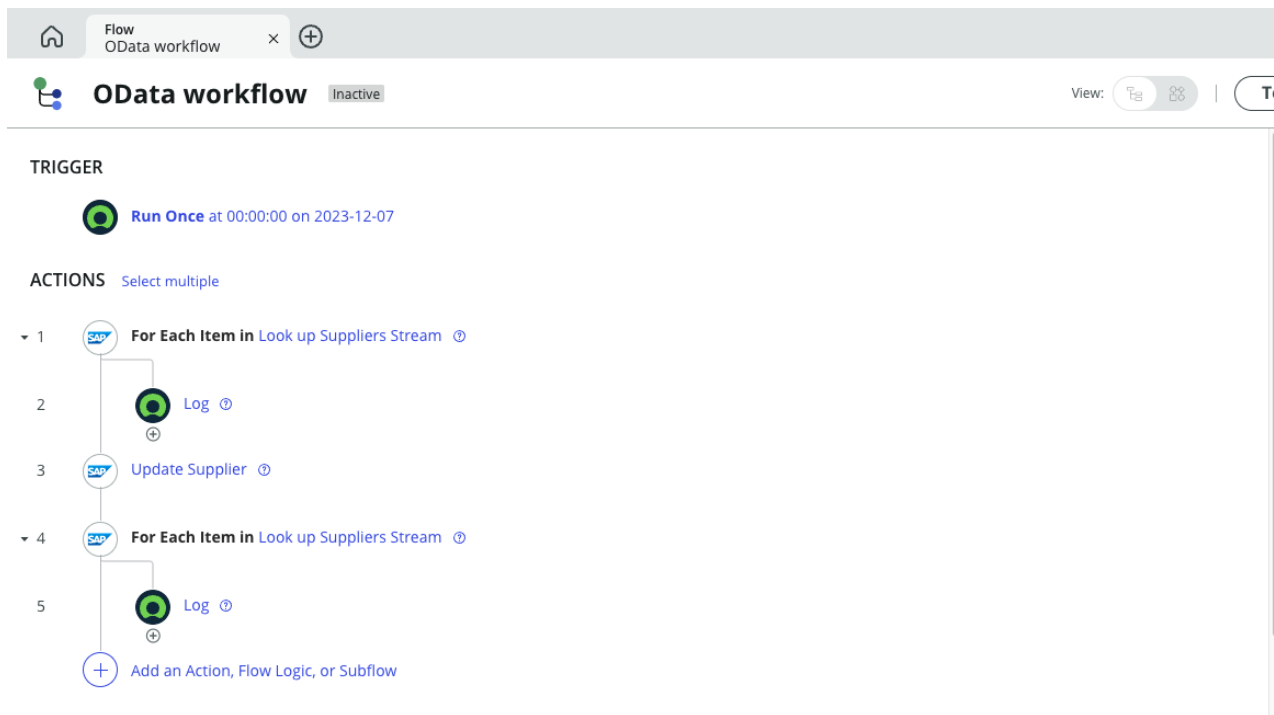
Test Case 1: Look Up Supplier and Update Supplier

ServiceNow will trigger a “Look up Supplier” and “Update Supplier” SAP S/4HANA OData actions to return a list of suppliers and update a single supplier. Both requests will go to OData endpoints that will perform actions in the SAP S/4Hana system. Response will then come back to ServiceNow.

Step 1 - ServiceNow: Workflow Configuration

This workflow will demonstrate that we are able to fetch a list of existing suppliers and update an existing supplier.

1. In Flow Designer create a new workflow “OData workflow” and add Look up Suppliers, Update Supplier along with log action.



1. Use default configuration for Look up Supplier
2. For log action use data pill from Look up Suppliers to log Supplier Name

OData workflow Inactive View: [Icons] T

TRIGGER

Run Once at 00:00:00 on 2023-12-07

ACTIONS [Select multiple](#)

▼ 1 **For Each Item in** [Look up Suppliers Stream](#)

2 **Log**

Action

Level

Message

[Delete](#) [Cancel](#) [Done](#)

[Update Supplier](#)

Status: Draft | Application: SAP S4 HANA OData Spoke

1. Configure following fields in Update Supplier action to change Customer for specific Supplier.

OData workflow Inactive View: [Icons] T

3 **Update Supplier**

Action

▼ Schema

* Supplier

Alternative Payee Account Number

Authorization Group

Customer

Payment is Blocked for Supplier ☐

Posting is Blocked ☐

Purchasing is Blocked ☐

Supplier Account Group

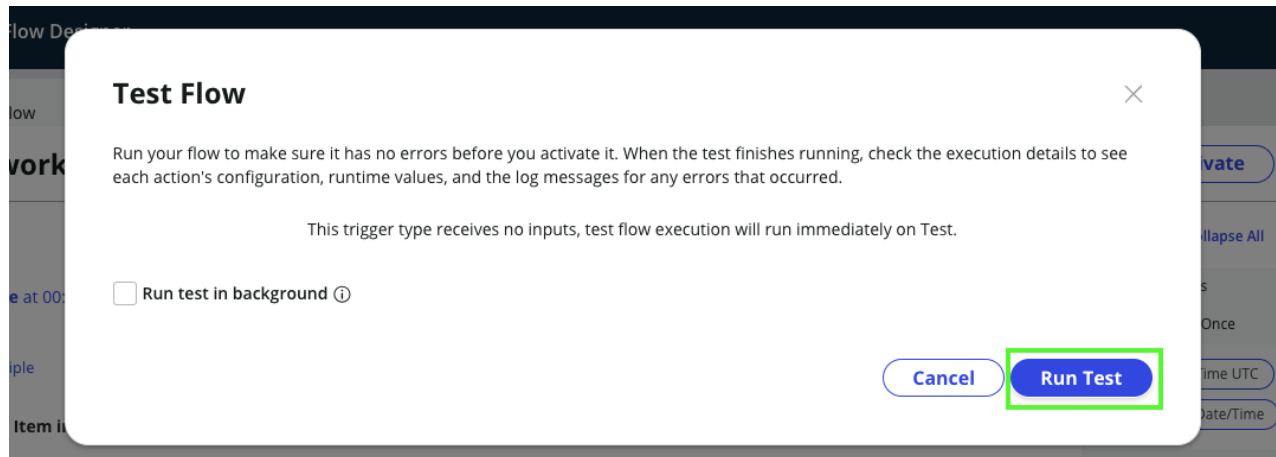
Birth Date

Status: Draft | Application: SAP S4 HANA OData Spoke

Note: 2nd use of Look up suppliers in workflow is optional.

Step 2 – ServiceNow: Execute Workflow

1. Open Flow Designer
2. Open workflow "OData Workflow"
3. Click on Test Workflow
4. Click on Run Test



Step 3 – ServiceNow: Inspect Request Look up Suppliers Query

1. After workflow execution is completed, open execution details
2. Open Step 2 execution details for Look up Suppliers action and inspect query

EXECUTION DETAILS OData workflow

Test Run - Completed [Open Flow](#) [Open Con](#)

Hide Action Details

	State	Start time
Query Parameters	<pre> [{"name":"\$top","value":"step > ... > pageSize", "attributes":{"omit_if_empty":true,"mandatory":true}}, {"name":"\$inlinecount","value":"allpages", "attributes":{"omit_if_empty":true,"mandatory":false}}, {"name":"\$skip","value":"step > ... > skip", "attributes":{"omit_if_empty":true,"mandatory":false}}, {"name":"\$select","value":"step > ... > Select", "attributes":{"omit_if_empty":true,"mandatory":false}}, {"name":"\$filter","value":"step > ... > Filter", "attributes":{"omit_if_empty":true,"mandatory":false}}, {"name":"\$orderby","value":"step > ... > Order By", "attributes":{"omit_if_empty":true,"mandatory":false}}, {"name":"\$expand","value":"step > ... > Expand", "attributes":{"omit_if_empty":true,"mandatory":false}}, {"name":"\$format","value":"json", "attributes":{"omit_if_empty":true,"mandatory":false}}] </pre>	Name/Values
Resource Path	/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier	String
Retry Policy	Default HTTP Retry Policy	Reference
Step Output Data		
VARIABLE NAME	RUNTIME VALUE	TYPE
Credential Value	*****	Password (2 Way Encrypted)
Error Code	0	String
Error Message		String
Response Body	{ "d": { "__count": "8", "results": [{ "__metadata": { "id": "http://ec2-18-170-34-13.eu-west-2.compute.amazon..." } }] } }	String
Response Headers	{ "set-cookie": ["sap-usercontext=sap-client=000; path=/", "SAP_SESSIONID_S4X_000=11vjelNSgL7hY0P8iIfjZ..."] }	String
Status Code	200	String
Step Status	{ "Step Status": { "code": 0, "message": "Success" } }	Object

Query (JSON) can be found in the resources directory.

Step 4 - ServiceNow: Inspect Request Update Supplier Request

1. After workflow execution is completed, open execution details
2. Open Step 2 execution details for Update Supplier action and inspect request

EXECUTION DETAILS

OData workflow

Test Run - Completed

Action Details

Enable Retry Policy	true	true
HTTP Method	patch	patch
Headers	[{"name":"X-CSRF-Token","attributes":{"omit_if_empty":false,"mandatory":false},"value":"i0OMLFm2aYPE..."}]	
Honor Duplicate Headers	true	true
Honor Duplicate Query Params	true	true
Override Base URL	false	
Override Default Policy for Alias	false	
Query Parameters	{}]	{}]
Request Body[Text]	{"PostingIsBlocked":false,"DeletionIndicator":false,"Customer":"0000100003","PaymentIsBlockedForSupp..."} payload	
Request Type	text	text
Resource Path	/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100003')	/sap/opu/odata/sap/API_BUSINESS_PARTNER/step ▶ ... ▶ Supplier '')
Retry Policy	Default HTTP Retry Policy ⓘ	
Save As Attachment	false	

Step Output Data

VARIABLE NAME	RUNTIME VALUE
Credential Value	*****
Error Code	0
Error Message	
Response Body	

Request payload (JSON) can be found in the resources directory.

Step 5 – SAP Execution

SAP Gateway Application Logs						
< SAP SAP Gateway Application Logs Exit 🔍 ? 🔗 Technical Information 📄 More						
External ID of the Log	Number	Altern ID	GSDO Type	Operation	Direct...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	1	/SAP/API_COMP...			Consum...	
> ■ No request ID has been provided	1	/SAP/API_COMP...			Consum...	
> ■ No request ID has been provided	1	/SAP/API_COMP...			Consum...	
> ■ No request ID has been provided	1	/SAP/API_COMP...			Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	1	/SAP/API_BUSIN...		update	Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	
> ■ No request ID has been provided	9	/SAP/API_BUSIN...		read	Consum...	

Step 6 – ServiceNow: Inspect Look up Suppliers Response Payload

1. After workflow execution is completed, open execution details
2. Open Step 2 execution details for Look up Suppliers action and inspect response body

EXECUTION DETAILS			State	Start time
Hide Action Details				
Credential Value	*****			Password (2 Way Encrypted)
Error Code	0			String
Error Message				String
Response Body	{"d":{"__count":"8","results":[{"__metadata":{"id":"http://ec2-18-170-34-13.eu-west-2.compute.amazon...			String
Response Headers	{"set-cookie":["sap-usercontext=sap-client=000; path=/","SAP_SESSIONID_S4X_000=1vjelN5gL7hY0P8iifjZ...			String
Status Code	200			String
Step Status	{"Step Status":{"code":0,"message":"Success"}}			Object

Object can be further inspected by clicking on value.

Viewing response_body [string]



Rendered HTML

Raw Text

Code

```
1 [{"d":{"__count":"8","results":[{"__metadata":{"id":"http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100000')","uri":"http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100000')","type":"API_BUSINESS_PARTNER.A_Supplier Type"},"Supplier":"100000","AlternativePayeeAccountNumber":"","AuthorizationGroup":"","CreatedByUser":"SRAMALINGAM","CreationDate":"","Date(1700611200000)":"","Customer":"","PaymentIsBlockedForSupplier":false,"PostingIsBlocked":false,"PurchasingIsBlocked":false,"SupplierAccountGroup":"KRED","SupplierFullName":"Hitachi Digital/W1 5DU","SupplierName":"Hitachi Digital","VATRegistration":"","BirthDate":null,"ConcatenatedInternationalLocNo":"00000000 00000000","DeletionIndicator":false,"FiscalAddress":"","Industry":"","InternationalLocationNumber1":"0","InternationalLocationNumber2":"0","InternationalLocationNumber3":"0","IsNaturalPerson":"","ResponsibleType":"","SuplrQtyInProcmtCertfnValidTo":null,"SuplrQualityManagementSystem":"","SupplierCorporateGroup":"","SupplierProcurementBlock":"","TaxNumber1":"","TaxNumber2":"","TaxNumber3":"","TaxNumber4":"","TaxNumber5":"","TaxNumberResponsible":"","TaxNumberType":"","SuplrProofOfDelivRlvtCode":"","BR_TaxIsSplit":false,"DataExchangeInstructionKey":"","to_SupplierCompany":{"__deferred":{"uri":"http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100000')/to_SupplierCompany"},"to_SupplierPurchasingOrg":{"__deferred":{"uri":"http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100000')/to_SupplierPurchasingOrg"},"to_SupplierText":{"__deferred":{"uri":"http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100000')/to_SupplierText"}}},"__metadata":{"id":"http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100001')","uri":"http://ec2-18-170-34-13.eu-west-2.compute.amazonaws.com:8000/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100001')","type":"API_BUSINESS_PARTNER.A_Supplier Type"},"Supplier":"100001","AlternativePayeeAccountNumber":"","AuthorizationGroup":"","CreatedByUser":"SRAMALINGAM","CreationDate":"","Date(1700611200000)":"","Customer":"","PaymentIsBlockedForSupplier":false,"PostingIsBlocked":false,"PurchasingIsBlocked":false,"SupplierAccountGroup":"KRED","SupplierFullName":"Hitachi Vantara/600009 bangalore","SupplierName":"Hitachi Vantara","VATRegistration":"","BirthDate":null,"ConcatenatedInternationalLocNo":"00000000 00000000","DeletionIndicator":false,"FiscalAddress":"","Industry":"","InternationalLocationNumber1":"0","InternationalLocationNumber2":"0","InternationalLocationNumber3":"0","IsNaturalPerson":"","ResponsibleType":"","SuplrQtyInProcmtCertfnValidTo":null,"SuplrQualityManagementSystem":"","SupplierCorporateGroup":"","SupplierProcurementBlock":"","TaxNumber1":"","TaxNumber2":"","TaxNumber3":""}]}
```

Close

Response (JSON) can be found in the resources directory.

Step 7 – ServiceNow: Inspect Update Supplier Response

1. After workflow execution is completed, open execution details
2. Open Step 2 execution details for Update Supplier action and inspect step output.

Headers	[{"name":"X-CSRF-Token","attributes":{"omit_if_empty":false,"mandatory":false},"value":"f0OMLFm2aYPE..."]	
Honor Duplicate Headers	true	true
Honor Duplicate Query Params	true	true
Override Base URL	false	
Override Default Policy for Alias	false	
Query Parameters		
Request Body[Text]	{ "PostingIsBlocked":false, "DeletionIndicator":false, "Customer":"0000100003", "PaymentIsBlockedForSupp... }	
Request Type	text	text
Resource Path	/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100003')	/sap/opu/odata/sap/API_BUSINESS_PARTNER/A_Supplier('100003')
Retry Policy	Default HTTP Retry Policy ⓘ	
Save As Attachment	false	

Step Output Data	
VARIABLE NAME	RUNTIME VALUE
Credential Value	*****
Error Code	0
Error Message	
Response Body	
Response Headers	{ "content-length":["0"], "sap-server":["true"], "sap-processing-info":["ODataBEP=,crp=,RAL=,st=,MedCac..."] }
Status Code	204
Step Status	{ "Step Status":{ "code":0, "message":"Success" } }

In this case, status code 204 is indicative of success.

Step 8 – Validation on SAP Side

One can validate the PO created from the response from servicenow and open it in SAP ME23N.

SAP Data Browser: Table LFA1 Select Entries 1																
Back (F3)	63	Q	↺	Check Table...	≡	≡	≡	≡	≡	≡	≡	≡	≡	≡	More	⌕
Cl.	Vendor	Cty	Name 1	Name 2	Name 3	Name 4	City	District	PO Box	PO ...	Postal...	Rg	SearchTer...	Street	Address	Name 1
000	0000100000	IN	Hitachi				banga...				600009	10	HITACHI		0000022965	HITACHI